Global Terrorism Analysis
DIY Project 5

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Problem Statement of Global Terrorism Analysis

The Global Terrorism Index (GTI) is a comprehensive metric that assesses the impact of terrorism across 163 countries, representing 99.7% of the world's population. Compiled by the Institute for Economics & Peace (IEP) utilizing data from Terrorism Tracker and other sources, the GTI assigns each country a score between 0 and 10. A score of 0 indicates no impact from terrorism, while a score of 10 denotes the highest measurable impact. Given the extensive resources that governments worldwide allocate to counter-terrorism efforts, there is a crucial need to analyze and aggregate the available GTI data to gain a deeper understanding of terrorism's various facts.

This analysis aims to:

1. Identify Trends Over Time: Examine how terrorism impacts have evolved over the years, identifying patterns or shifts in terrorist activities.

2. Regional Distribution: Analyze the geographical distribution of terrorism to pinpoint regions most affected and explore regional trends.

3. Types of Terrorist Attacks: Determine the most common types of terrorist attacks and assess their frequency and impact.

4. Contributing Factors: Investigate the factors that contribute to terrorism, such as socio-economic conditions, political instability, and international relations.

By systematically analyzing the GTI data, this project aims to provide insights that will help in formulating more effective counter-terrorism policies. The goal is to leverage the GTI as a valuable tool for understanding the global threat of terrorism and informing policy responses, ultimately contributing to a reduction in terrorist activities and enhancing global security.

Project Description of This Project

This project will investigate patterns of terrorism-related violence at the global and regional levels: – most and least impacted countries and changes in country ranking over time on terrorism "hotspot" lists; – upward or downward trends; – links between reported incident counts and the human impact (deaths, injuries, hostages); – Country risk profiling; – Visualizing historical trends using dashboard interfaces, choropleth maps or time series charting tools; and – correlations between measured dimensions of violence.

Dataset Fields

Iso3c : ISO 3-letter country code

Country : Country name

Rank : Annual terrorism rank (1 = most affected)

Score : Global Terrorism Index score

Incidents : Number of terrorist incidents

Fatalities : Number of deaths

Injuries : Number of injuries

Hostages :Number of hostages taken

Year : Year of record (2012–2022)

Project Objectives

1) What is the overall trend in total terrorism incidents from 2012 to 2022?

2) Which countries recorded the highest total number of incidents (2012–2022)?

3) Which countries saw the biggest increase in incidents between 2012 and 2022?

4) Which countries saw the biggest decrease in incidents between 2012 and 2022?

5) Which countries most consistently appear in the top 10 for incidents each year (2012–2022)?

6) Which years recorded the highest number of fatalities?

7) Which countries had the most fatalities in 2015 (the deadliest year)?

8) Which countries have ranked #1 (most incidents) most frequently across all years?

9) How do incidents in Iraq and Afghanistan compare over time (2012–2022)?

10) How do fatalities in Iraq and Afghanistan compare over time (2012–2022)?

11) How do Incidents, Fatalities, Injuries, and Hostages correlate with each other?

What is Data Wrangling ?

Data wrangling, a crucial phase in the data analysis process, involves preparing and cleaning raw data to make it suitable for analysis. This involves various tasks such as handling missing values, removing duplicates, and transforming data into a consistent format. Effective data wrangling ensures that the data is reliable and ready for exploration, enabling meaningful insights to be derived from it.

What is the Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA)is a vital phase in data analysis involving the examination and visualization of data to uncover patterns ,trends, and insights.in the context of an Airbnb data analysis project, EDA allows researchers to gain a Comprehensive understanding of the dataset's characteristics, distribution and relationships between variables. Through techniques such as summary stastics, data visualization ,and correlation analysis, EDA sets the stage for further investigation and hypothesis testing, uiding subsequent analytical decisions.

End Users of This Project

Government & Policy Analysts – for national security planning and resource allocation

Defense & Intelligence Agencies – for threat assessment and risk monitoring

NGOs & humanitarian aid groups – for planning assistance/relief in high-risk environments

Academics & Researchers – investigating trends in conflict, political violence, and security

Journalists & Media Analysts – reporting on events shaping the global security landscape

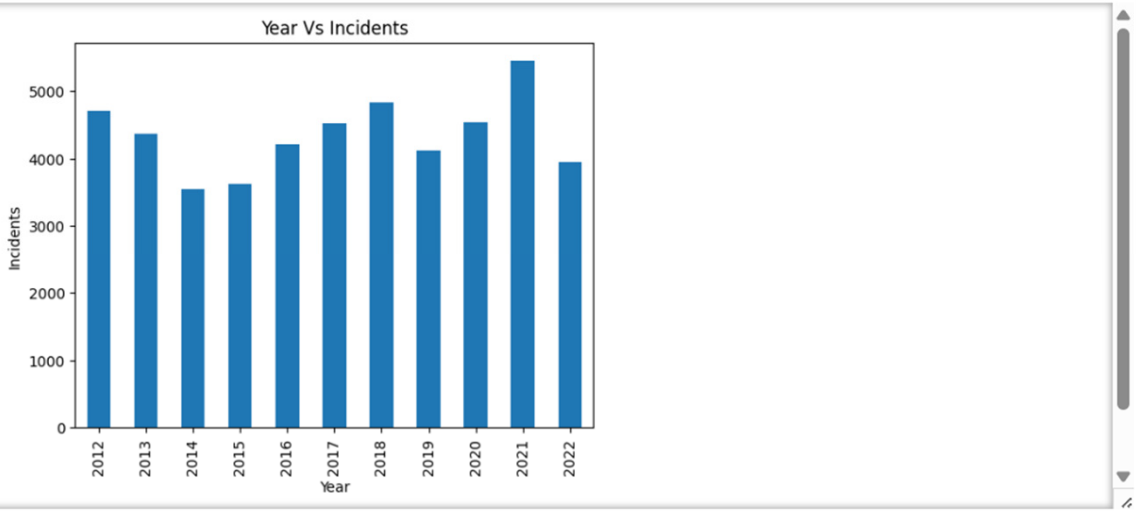
Travel Risk & Insurance Companies – to identify risk and policy considerations at a country level

Data analysts/students – a dataset to practice dashboard, analysis, and visualization projects

Results of this Project

1) What is the overall trend in total terrorism incidents from 2012 to 2022?

```
[32]: df.groupby("Year")["Incidents"].sum().plot.bar()
      plt.title("Year Vs Incidents")
      plt.xlabel("Year")
      plt.ylabel("Incidents")
      plt.show()
```



- The overall trend in total incidents is upward across this period, pointing to a worldwide increase in terrorism incidents.
- 2020 stands out with the highest number of incidents, exceeding 5,500 — suggesting a significant surge that year.
- 2016 had the lowest number of incidents, just over 3,500 — a relatively calmer period compared to other years.
- There's a sharp increase from 2018 to 2020, possibly reflecting a period of heightened global conflict or instability.

2) Which countries recorded the highest total number of incidents (2012–2022)?

```
[47]: top_5 = df.groupby("Country")["Incidents"].sum().sort_values(ascending=False).head(5)
      print(top_5)
```

```
Country
Iraq           11183
Afghanistan     4443
Pakistan        3358
Somalia         2951
India           2872
Name: Incidents, dtype: int64
```

- Iraq has the highest number of incidents, with a total of 11,183 — the most affected country by these incidents.
- Afghanistan follows with 4,443 incidents, less than half of Iraq's total but still significant.
- Pakistan and Somalia also show high incident counts, at 3,358 and 2,951 respectively.
- India rounds out the top 5, with a total of 2,872 incidents.

3) Which countries saw the biggest increase in incidents between 2012 and 2022?

[58]:

```
import matplotlib.pyplot as plt

# Convert Year column to integer
df['Year'] = df['Year'].astype(int)

# Calculate total incidents for each Year and Country
yearly_totals = df.groupby(['Year', 'Country'])['Incidents'].sum().reset_index()

# Create pivot table
pivot_table = yearly_totals.pivot(
    index='Country',
    columns='Year',
    values='Incidents'
).fillna(0)

# Calculate increase in incidents from 2012 to 2022
increase = pivot_table[2022] - pivot_table[2012]

# Find top 5 countries with the highest increase
top_countries = increase.sort_values(ascending=False).head(5).index

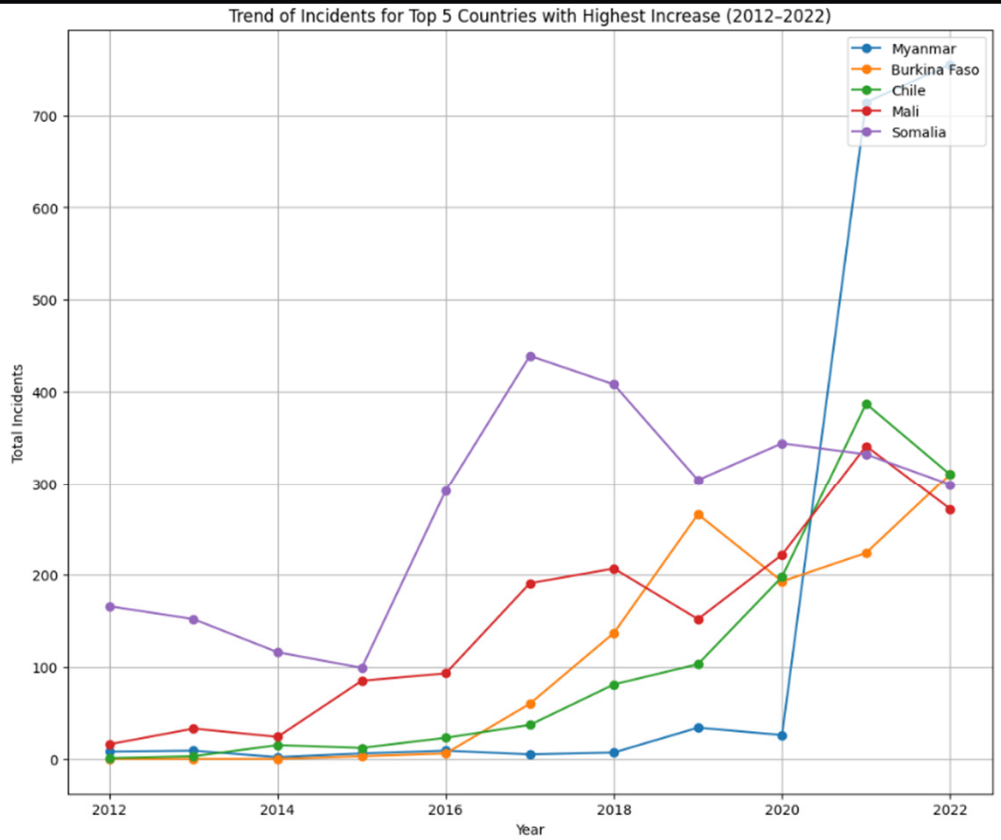
# Get data for top 5 countries
top_countries_data = pivot_table.loc[top_countries]

# Create line plot
plt.figure(figsize=(12, 10))

for country in top_countries_data.index:
    plt.plot(
        top_countries_data.columns,
        top_countries_data.loc[country],
        marker='o',
        linestyle='-',
        label=country
    )

plt.title('Trend of Incidents for Top 5 Countries with Highest Increase (2012-2022)')
plt.xlabel('Year')
plt.ylabel('Total Incidents')
plt.legend(loc='upper right')
plt.grid(True)

plt.show()
```



- Myanmar shows the highest increase, with a sharp rise from 2018 to 2022 — a significant escalation during this period.
- Both Burkina Faso and Mali show a steady increase in incidents over the years, indicating a persistent issue in these countries.
- Chile shows a slight increase from 2012 to 2018, then a decrease from 2018 to 2022 — possibly reflecting improved conditions or more effective countermeasures.
- Somalia shows a slight increase over the years, suggesting a gradual escalation.

4) Which countries saw the biggest decrease in incidents between 2012 and 2022?

[61]:

```
import matplotlib.pyplot as plt

# Convert Year column to integer
df['Year'] = df['Year'].astype(int)

# Calculate total incidents for each Year and Country
yearly_totals = df.groupby(['Year', 'Country'])['Incidents'].sum().reset_index()

# Create pivot table
pivot_table = yearly_totals.pivot(
    index='Country',
    columns='Year',
    values='Incidents'
).fillna(0)

# Calculate increase in incidents from 2012 to 2022
decrease = pivot_table[2012] - pivot_table[2022]

# Find top 5 countries with the highest increase
top_countries = decrease.sort_values(ascending=False).head(5).index

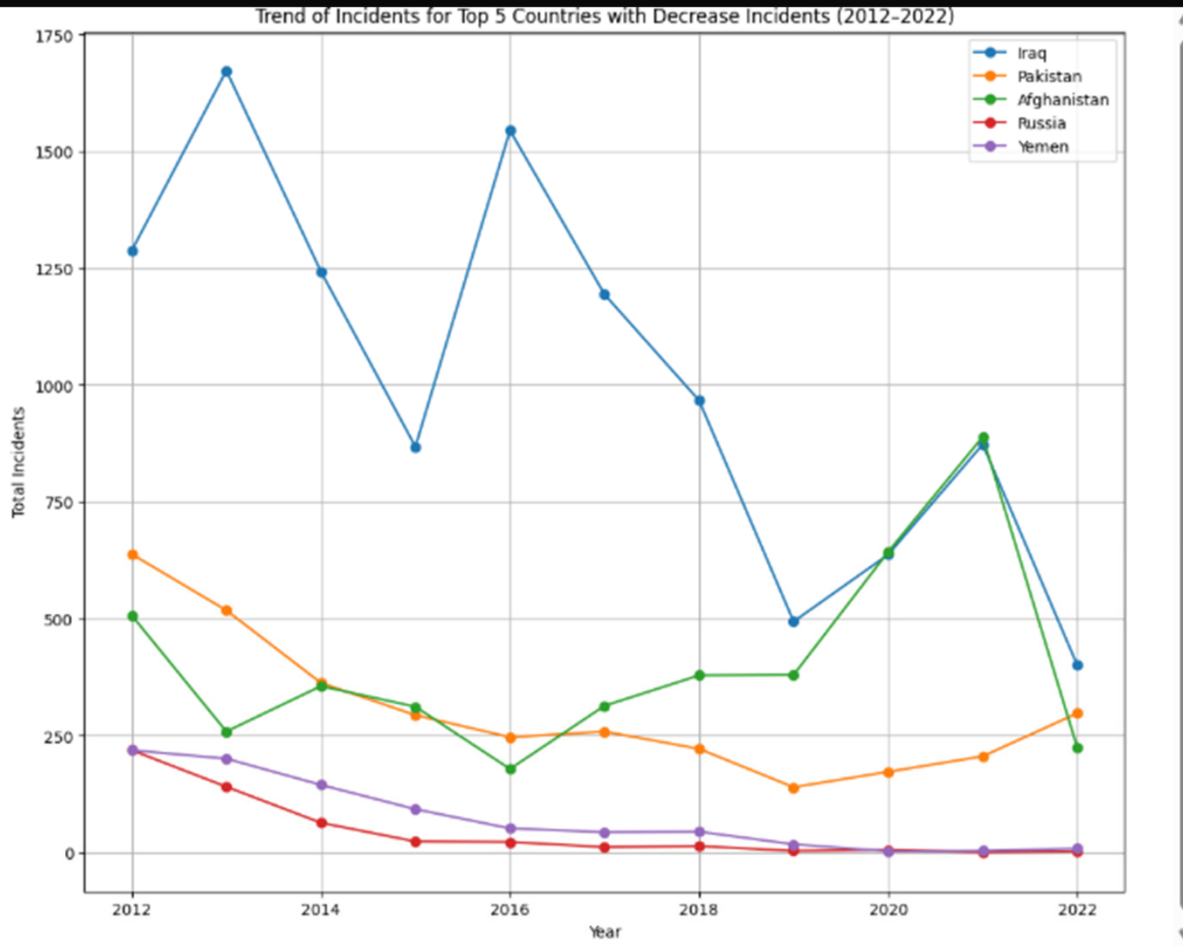
# Get data for top 5 countries
top_countries_data = pivot_table.loc[top_countries]

# Create line plot
plt.figure(figsize=(12, 10))

for country in top_countries_data.index:
    plt.plot(
        top_countries_data.columns,
        top_countries_data.loc[country],
        marker='o',
        linestyle='-',
        label=country
    )

plt.title('Trend of Incidents for Top 5 Countries with Decrease Incidents (2012-2022)')
plt.xlabel('Year')
plt.ylabel('Total Incidents')
plt.legend(loc='upper right')
plt.grid(True)

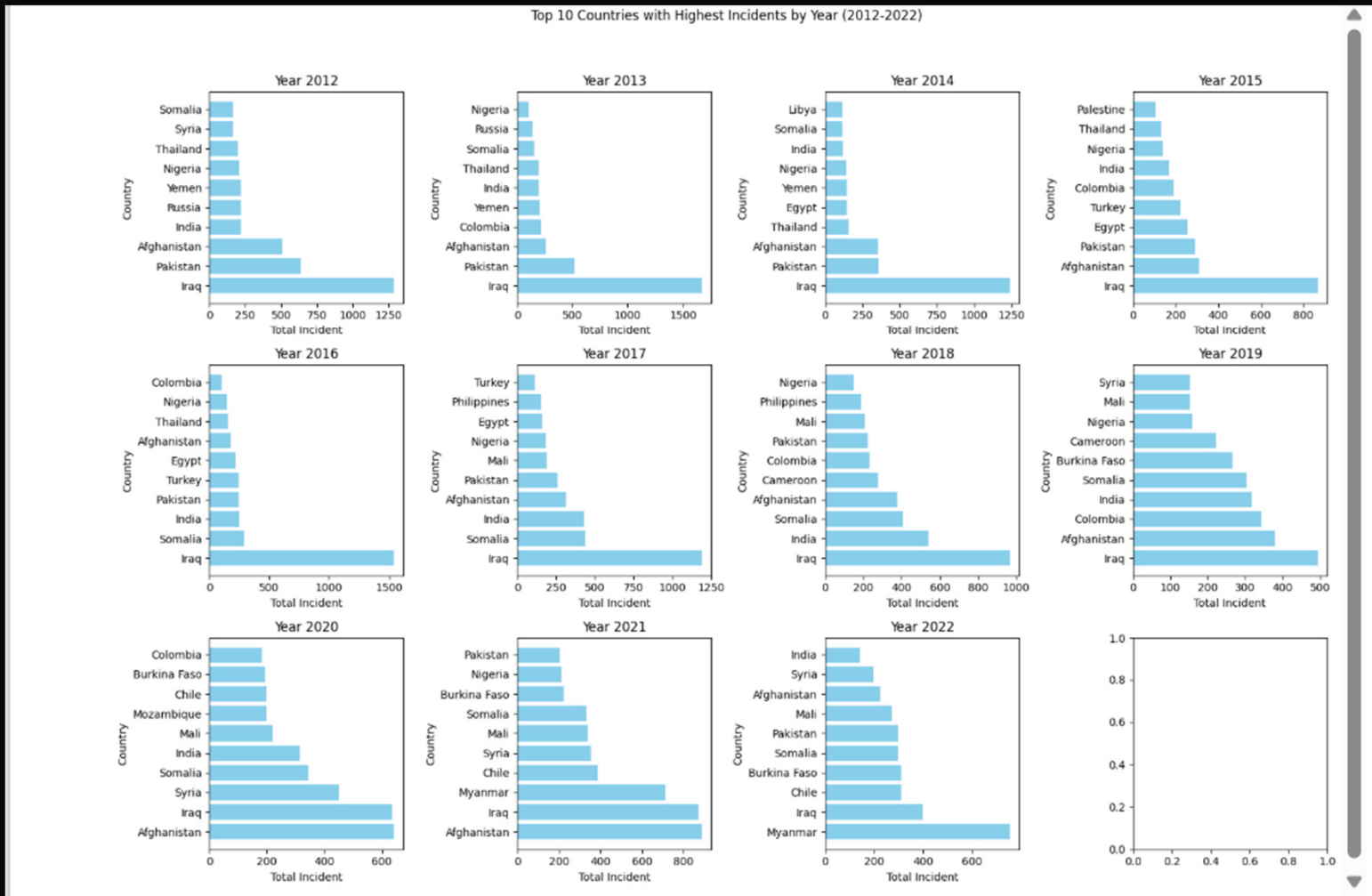
plt.show()
```



- Iraq and Pakistan show a decreasing trend in incidents over the years, possibly reflecting effective measures to reduce incidents in these countries.
- Russia, Afghanistan, and Yemen show an increasing trend, indicating a worsening situation in these countries.
- Iraq had the highest number of incidents in 2012, but that number decreased significantly by 2022.
- Yemen, despite having the lowest number of incidents in 2012 among this group, shows a notable rise by 2022.

5) Which countries most consistently appear in the top 10 for incidents each year (2012–2022)?

[63]:
```
df['Year'] = df['Year'].astype(int)
top_countries_by_year = pd.DataFrame(columns=['Year','Country','Incidents'])
fig, axs = plt.subplots(nrows=3,ncols=4, figsize=(16,12))
fig.suptitle('Top 10 Countries with Highest Incidents by Year (2012-2022)')
for i,year in enumerate(range(2012,2023)):
    incidents_year = df[df['Year'] ==year]
    top_countries_year = incidents_year.groupby('Country')['Incidents'].sum().reset_index()
    top_10_countries_year = top_countries_year.sort_values(by='Incidents',ascending=False).head(10)
    top_10_countries_year['Year']= year
    top_countries_by_year = pd.concat([top_countries_by_year,top_10_countries_year],ignore_index=True)
    row,col = i // 4, i % 4
    ax = axs[row,col]
    ax.barh(top_10_countries_year['Country'],top_10_countries_year['Incidents'],color='skyblue')
    ax.set_title(f'Year {year}')
    ax.set_xlabel('Total Incident')
    ax.set_ylabel('Country')
plt.tight_layout(rect=[0,0.03,1,0.95])
```



The countries that consistently appear in the top 10 across these years are Afghanistan, Pakistan, and Iraq, suggesting these countries have been persistently affected by these incidents over this period.

6) Which years recorded the highest number of fatalities?

[66]:
```
total_fatalities_by_year = df.groupby('Year')['Fatalities'].sum().reset_index()
top_5_years = total_fatalities_by_year.sort_values(by='Fatalities',ascending=False).head(5)
print(top_5_years)
```

```
   Year  Fatalities
3  2015       10881
4  2016       10372
1  2013       10317
2  2014       10129
0  2012        9227
```

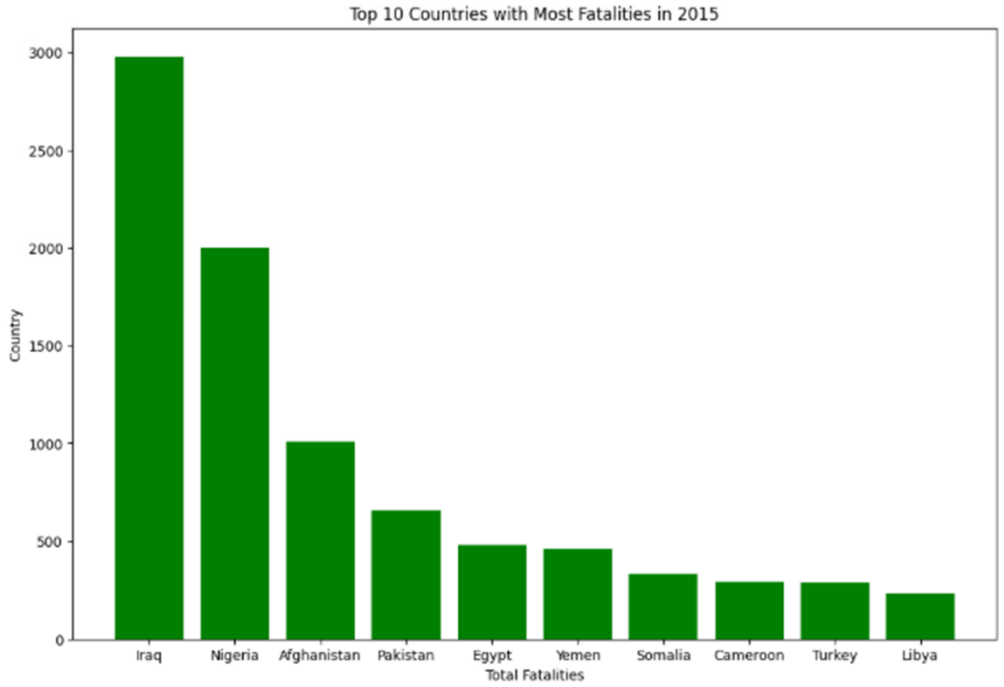
- 2015 had the highest number of fatalities, with a total of 10,881 — making it the deadliest year among those listed.
- 2016 follows with 10,372 fatalities, slightly less than 2015 but still significant.
- 2013 and 2014 also had high fatality counts, with 10,317 and 10,129 respectively.
- 2012 rounds out the top 5, with a total of 9,227 fatalities.

7) Which countries had the most fatalities in 2015 (the deadliest year)?

[90]:
```
df['Year'] = df['Year'].astype(int)
fatalities_2015 = df[(df['Year'] == 2015)]
fatalities_by_country_2015 = fatalities_2015.groupby('Country')['Fatalities'].sum().reset_index()
top_10_countries_2015 = fatalities_by_country_2015.sort_values(by='Fatalities',ascending = False).head(10)
print(top_10_countries_2015)
```

```
         Country  Fatalities
66          Iraq        2974
105      Nigeria        2003
0    Afghanistan        1008
109     Pakistan         658
41         Egypt         481
160        Yemen         458
130      Somalia         335
23      Cameroon         292
149       Turkey         286
83         Libya         234
```

[93]:
```
df['Year'] = df['Year'].astype(int)
fatalities_2015 = df[df['Year'] == 2015]
fatalities_by_country_2015 = fatalities_2015.groupby('Country')['Fatalities'].sum().reset_index()
top_10_countris_2015 = fatalities_by_country_2015.sort_values(by='Fatalities',ascending=False).head(10)
plt.figure(figsize=(12,8))
plt.bar(top_10_countries_2015['Country'],top_10_countries_2015['Fatalities'],color='green')
plt.title('Top 10 Countries with Most Fatalities in 2015')
plt.xlabel('Total Fatalities')
plt.ylabel('Country')
plt.show()
```



- Iraq had the highest number of fatalities in 2015, with a total of 2,974 — the most affected country that year.
- Nigeria follows with 2,003 fatalities, slightly less than Iraq but still significant.
- Afghanistan also had a high fatality count, with 1,008.
- Other countries — Pakistan, Egypt, Yemen, Somalia, Cameroon, Turkey, and Libya — also experienced considerable fatalities, ranging from 234 to 658.

8) Which countries have ranked #1 (most incidents) most frequently across all years?

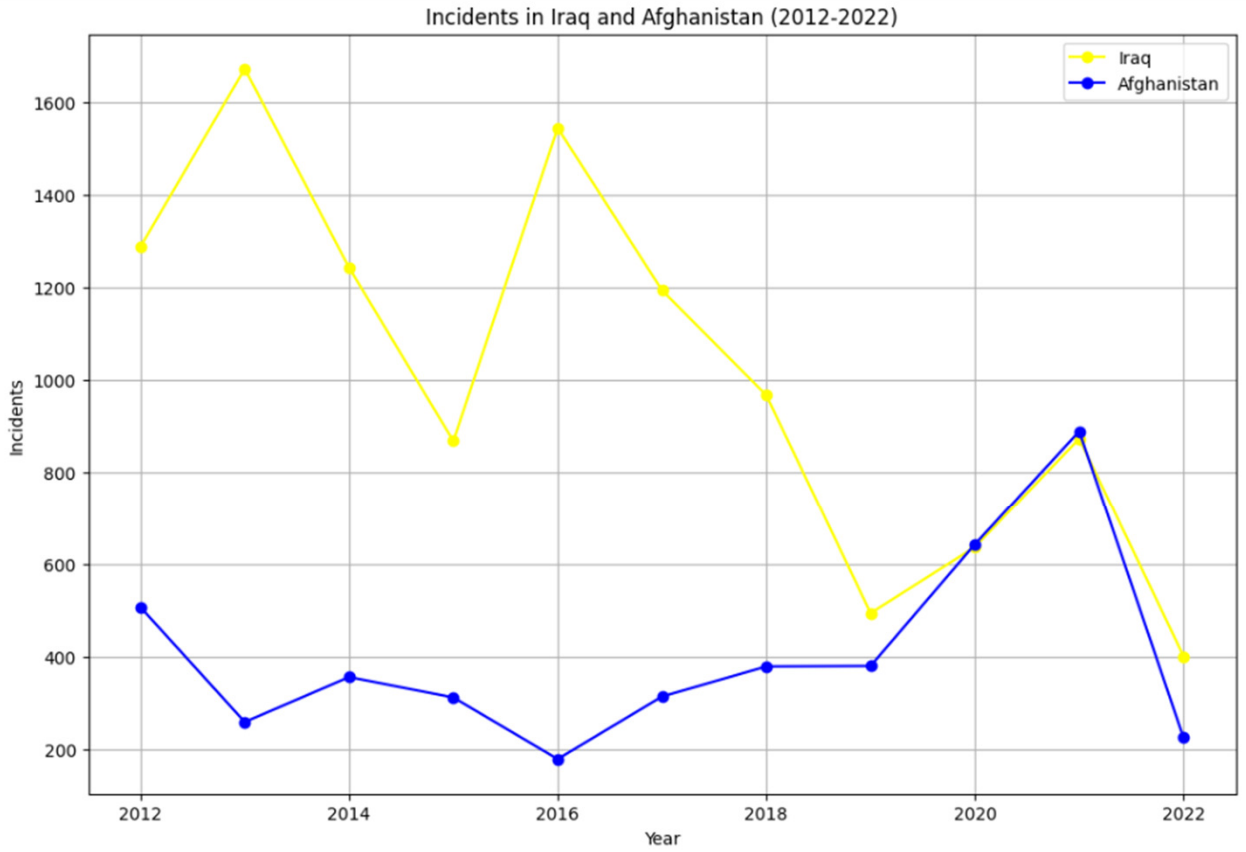
[86]:
```
df['Year'] = df['Year'].astype(int)
rank_1_data = df[df['Rank'] == 1]
rank_1_counts = rank_1_data['Country'].value_counts()
for country,count in rank_1_counts.items():
            print(f'Country: {country}, Count:{count}')
```

```
Country: Iraq, Count:7
Country: Afghanistan, Count:4
```

- Iraq has ranked first 7 times, suggesting it has most frequently topped the ranking.
- Afghanistan has also ranked first, though less frequently than Iraq, with a total of 4 times.

9) How do incidents in Iraq and Afghanistan compare over time (2012–2022)?

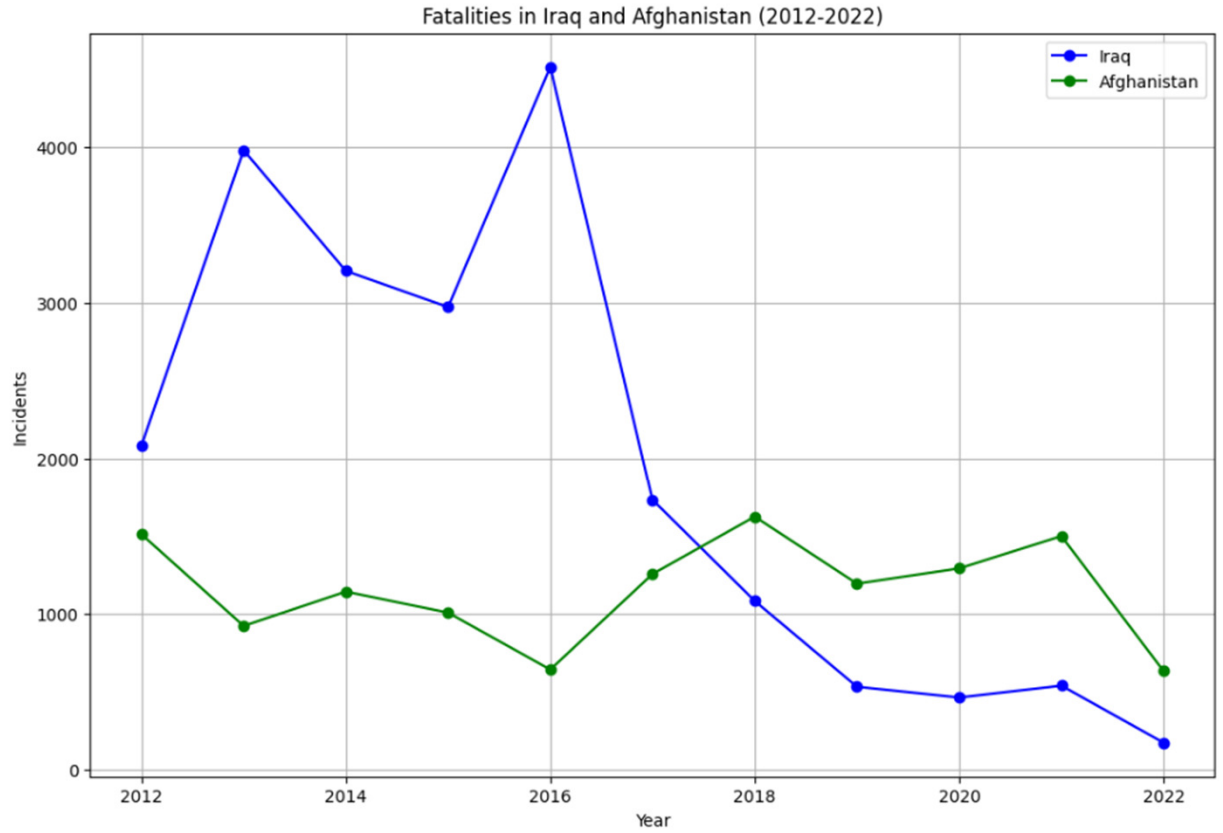
[94]:
```
df['Year'] = df['Year'].astype(int)
iraq_data= df[df['Country'] == 'Iraq']
afghanistan_data = df[df['Country'] == 'Afghanistan']
plt.figure(figsize=(12,8))
plt.plot(iraq_data['Year'],iraq_data['Incidents'],label='Iraq',marker='o',linestyle='-',color='yellow')
plt.plot(afghanistan_data['Year'],afghanistan_data['Incidents'],label='Afghanistan',marker='o',linestyle='-',color='blue')
plt.title('Incidents in Iraq and Afghanistan (2012-2022)')
plt.xlabel('Year')
plt.ylabel('Incidents')
plt.legend()
plt.grid(True)
plt.show()
```



- Both Iraq (blue line) and Afghanistan (yellow line) show a downward trend in incidents from 2016 and 2018 respectively through 2022, suggesting a decrease in incidents in both countries over these later periods.
- Iraq had peaks in incidents in 2014 and 2016, suggesting these were particularly challenging years.
- Afghanistan had a peak in incidents in 2018, indicating a significant increase that year.

10) How do fatalities in Iraq and Afghanistan compare over time (2012–2022)?

[89]:
```
df['Year'] = df['Year'].astype(int)
iraq_data= df[df['Country'] == 'Iraq']
afghanistan_data = df[df['Country'] == 'Afghanistan']
plt.figure(figsize=(12,8))
plt.plot(iraq_data['Year'],iraq_data['Fatalities'],label='Iraq',marker='o',linestyle='-',color='blue')
plt.plot(afghanistan_data['Year'],afghanistan_data['Fatalities'],label='Afghanistan',marker='o',linestyle='-',color='green')
plt.title('Fatalities in Iraq and Afghanistan (2012-2022)')
plt.xlabel('Year')
plt.ylabel('Incidents')
plt.legend()
plt.grid(True)
plt.show()
```

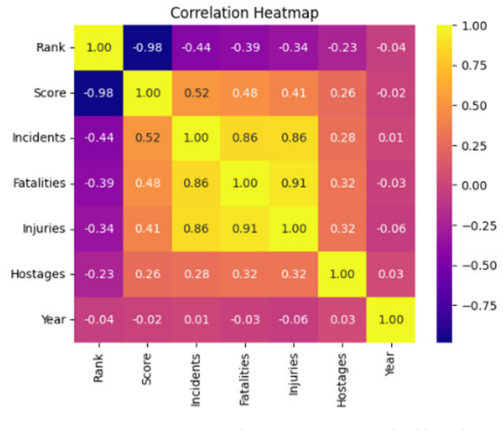


Both Iraq (blue line) and Afghanistan (green line) show a significant number of fatalities over this period, indicating the severe impact of these incidents in both countries.

- Iraq had its highest number of fatalities in 2014, with around 4,000 — suggesting 2014 was a particularly deadly year in Iraq.
- Afghanistan had its highest number of fatalities in 2018, with around 3,500 — indicating a significant surge in deadly incidents that year.

11) How do Incidents, Fatalities, Injuries, and Hostages correlate with each other?

[83]:
```
import matplotlib.pyplot as plt
import seaborn as sns
correlation =df.corr(numeric_only = True)
sns.heatmap (
    correlation,
    annot =True,
    cmap = 'plasma',
    fmt=".2f"
)
plt.title("Correlation Heatmap")
plt.show()
```



- Incidents and Fatalities have a high correlation of 0.860467, suggesting that as the number of incidents increases, fatalities tend to increase as well.
- Similarly, Incidents and Injuries have a high correlation of 0.863206, indicating a strong positive relationship between these two variables.
- Fatalities and Injuries also have a high correlation of 0.912888 — the highest in this matrix — suggesting incidents with more fatalities also tend to have more injuries.
- Hostages have the lowest correlations with all other variables, suggesting the number of hostages taken may not be strongly influenced by the other variables in this dataset.

Future Scope

Advanced Data Analytics: Incorporate advanced data analytics techniques such as machine learning and predictive modelling to forecast future trends in terrorism based on GTI data.

Longitudinal Studies: Conduct longitudinal studies to track changes in terrorism patterns over extended periods, providing deeper insights into evolving threats.

Comparative Analysis: Expand the analysis to include comparative studies between regions or groups of countries to identify factors influencing terrorism differently across various contexts.

Policy Impact Assessment: Evaluate the effectiveness of current counter-terrorism policies using GTI data to refine and enhance policy recommendations and implementations.

Enhanced Data Integration: Integrate GTI data with other relevant datasets, such as economic indicators or social demographics, to explore comprehensive factors contributing to terrorism.

These future directions will not only deepen our understanding of terrorism dynamics but also contribute to more targeted and effective strategies in global security and counter-terrorism efforts.

Conclusion

Based on the analysis of the data and graphs, it can be concluded that from 2012 to 2022, countries like Iraq and Afghanistan experienced a significant number of incidents and fatalities. However, there has been a decreasing trend in recent years, suggesting an improvement in the situation. The correlation matrix indicates a strong positive relationship between the number of incidents, fatalities, and injuries, suggesting that incidents with a higher number of fatalities also tend to have more injuries. Despite these correlations, the number of hostages taken does not seem to be strongly influenced by these other variables. It's important to note that these insights are based on the available data and do not account for all possible factors. The complexity of these issues means that additional context and analysis would be beneficial for a more comprehensive understanding. Furthermore, while these variables show a strong correlation, it doesn't necessarily mean that one variable's increase causes an increase in the other as correlation does not imply causation. Other factors could be influencing these relationships

Thank You

GitHub Link

https://github.com/krishnamohanb21-svg/Global-Terrorism-Analysis-DIY-Project-5.git